

BUSINESS REQUIREMENTS DOCUMENT

Hospital Emergency Room Analytics Dashboard

Power BI | Data Analytics Project

Document Version	v1.0
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Project Name	Hospital Emergency Room Dashboard
Tool	Microsoft Power BI Desktop
Data Source	Hospital ER Excel Dataset (April 2023 – October 2024)
Status	Draft — For Stakeholder Review

1. Executive Summary

This Business Requirements Document (BRD) outlines the complete functional, analytical, and technical requirements for the Hospital Emergency Room Analytics Dashboard, developed in Microsoft Power BI. The dashboard is built to support hospital administrators, clinical leads, and operational stakeholders in making data-driven decisions to optimize emergency room (ER) performance, improve patient experience, and allocate resources efficiently.

The solution covers a 19-month period (April 2023 – October 2024) and encompasses 9,216 unique patient records. It delivers four purpose-built dashboard views — Monthly View, Consolidated View, Patient Details, and Key Takeaways — each designed to address a distinct analytical need.

2. Project Scope

2.1 In Scope

- Design and development of 4 Power BI dashboard views
- Data connection, cleaning, modeling, and transformation from the Excel source
- Creation of DAX measures and calculated columns
- KPI cards with sparklines for patient volume, wait time, satisfaction, and referrals
- Visual development: bar charts, column charts, donut charts, heatmap matrices, and data grids
- Descriptive analysis and key insights generation for stakeholder consumption
- Time intelligence using a dedicated Date Table

2.2 Out of Scope

- Real-time or live data streaming from hospital systems
- Integration with Electronic Health Records (EHR) or hospital management software

- Predictive or machine learning models
- Mobile-optimized dashboard views
- Patient-identifiable data — dataset is anonymized for privacy compliance

3. Business Objectives

The primary objectives driving this dashboard initiative are:

- Track and monitor total ER patient volume on a daily, monthly, and cumulative basis
- Reduce average patient wait times by identifying operational bottlenecks and peak hours
- Measure and improve patient satisfaction scores through trend analysis
- Understand referral patterns across departments to support resource planning
- Analyze patient demographics (age, gender, race) to identify healthcare access disparities
- Provide a single source of truth for ER performance reporting to all stakeholders
- Enable month-over-month and period-over-period comparisons for strategic planning

4. Stakeholders

Stakeholder	Role	Interest
Hospital Administrator	Executive Sponsor	Overall ER performance, cost efficiency, and patient outcomes
Clinical Lead / ER Head	Primary User	Wait times, referral rates, timeliness, and patient satisfaction
Operations Manager	Primary User	Staffing optimization, peak hours, and admission patterns
Data Analyst	Developer / Analyst	Data modeling, DAX, dashboard design, and insights generation
HR / Workforce Team	Secondary User	Staffing demand by day and hour for scheduling decisions
Quality & Compliance	Secondary User	Patient satisfaction, admission flags, and demographic equity

5. Data Source & Field Definitions

The dataset is sourced from the Hospital Emergency Room Excel file covering April 2023 through October 2024. It contains 9,216 unique patient records across the following fields:

Field Name	Description & Business Purpose
Patient ID	A unique alphanumeric code assigned to each patient. Primary identifier for tracking individual cases while ensuring privacy in data analysis.
Patient Admission Date	The date and time the patient was admitted to the ER. Used to analyze peak admission times, seasonal trends, and emergency response rates.

Patient First Initial	The first letter of the patient's first name, used for anonymization to comply with privacy regulations (HIPAA / GDPR).
Patient Last Name	The patient's last name. Used for case identification while maintaining anonymization standards in aggregated analysis.
Patient Gender	Gender identity recorded as Male, Female, or Not Confirmed. Used for demographic studies and healthcare disparity analysis.
Patient Age	Numerical age at time of admission. Critical for age-group analysis and identifying which groups most frequently visit the ER.
Patient Race	Self-reported racial or ethnic identity. Essential for studying healthcare access, outcomes, and disparities across groups.
Department Referral	The department the patient was referred to (e.g., General Practice, Orthopedics, Cardiology). Used for resource allocation analysis.
Patient Admin Flag	Binary field (True/False). True = admitted for further care; False = treated and released or referred to another facility.
Patient Satisfaction Score	Score on a scale of 1–10 capturing the patient's evaluation of their ER experience. Used for service quality analysis.
Patient Wait Time	Time elapsed from patient arrival to first medical staff contact. Critical for operational efficiency and patient experience analysis.
Patients CM (Case Manager)	The individual or team coordinating patient care. Used to analyze workload distribution and case management outcomes.

6. KPI Requirements

The following four KPIs form the core performance metrics of the dashboard. Each is displayed as a card visual with an area sparkline showing daily trends.

KPI Name	Description	Visual Type
No. of Patients	Total distinct patient count visiting the ER. Measured using DISTINCTCOUNT on Patient ID. Identifies peak days and seasonal volume trends.	KPI Card + Area Sparkline
Avg Wait Time	Average time (in minutes) patients wait before being attended to by medical staff. Highlights operational bottlenecks and days requiring adjustment.	KPI Card + Area Sparkline
Patient Satisfaction Score	Average satisfaction score (1–10 scale) per day. Dips correlate with high wait times and peak hour stress on staff.	KPI Card + Area Sparkline
No. of Patients Referred	Count of patients referred to specialist departments. Helps identify departments with high referral volumes needing additional resources.	KPI Card + Area Sparkline

7. Dashboard Requirements

The solution consists of four dashboard views, each serving a distinct analytical purpose:

Dashboard	Objective	Charts / Visuals to Develop
Dashboard 1 Monthly View	Monitor KPIs and trends on a month-by-month basis. Filtered by Month Name and Year slicers.	Patient Admission Status (table), Age Group Distribution (column chart), Department Referrals (bar chart), Timeliness — % seen within 30 mins (donut), Gender Distribution (donut), Race Demographics (bar chart), Patients by Day & Hour (heatmap matrix)
Dashboard 2 Consolidated View	Provide a holistic summary of ER performance across a customizable date range for broader trend analysis.	All Monthly View metrics aggregated over a selectable date range. Enables period-over-period comparison and trend identification across months and quarters.
Dashboard 3 Patient Details	Offer granular patient-level data to enable detailed analysis and individual case review.	Data grid table displaying: Patient ID, Full Name, Gender, Age, Admission Date, Race, Wait Time, Department Referral, Admission Status
Dashboard 4 Key Takeaways	Summarize findings across all dashboards with actionable insights for stakeholders.	Descriptive analysis text covering all KPIs, demographic findings, peak period analysis, referral patterns, admission trends, and recommendations

8. Calculated Columns & DAX Measures

8.1 DAX Measures

Measure Name	DAX Formula
No. of Patients	DISTINCTCOUNT('Hospital ER_Data'[Patient Id])
Avg Wait Time	AVERAGE('Hospital ER_Data'[Patient Waittime])
Patient Satisfaction Score	AVERAGE('Hospital ER_Data'[Patient Satisfaction Score])
No. of Patients Referred	CALCULATE([No. of Patients], 'Hospital ER_Data'[Department Referral] <> "None")

8.2 Calculated Columns

Column Name	Table	Logic / Purpose
Admission Status	Hospital ER_Data	IF([Patient Admission Flag] = TRUE, "Admitted", "Not Admitted") — converts binary flag to readable label
Age Group	Hospital ER_Data	SWITCH(TRUE(), ...) — groups Patient Age into 10-year bands: 0-9, 10-19, 20-29 ... 70-79

Waittime Interval	Hospital ER_Data	SWITCH(TRUE(), ...) — groups Admission Time into 2-hour slots: 00-02, 03-04 ... 23-24
Month & Year	Date Table	FORMAT([Date], "MMM YYYY") — creates readable month-year label for trend axis

9. Project Development Steps

The following steps define the end-to-end development workflow for this Power BI solution:

1. **Requirement Gathering / Business Requirements**
2. Data Walkthrough — understand structure, fields, and data quality
3. Data Connection — connect Power BI to the Excel data source
4. Data Cleaning / Quality Check — handle nulls, duplicates, and formatting issues
5. Data Modeling — build relationships between Date Table and Hospital ER_Data
6. Data Processing — create calculated columns (Age Group, Admission Status, Waittime Interval)
7. DAX Calculations — build all measures for KPIs
8. Dashboard Layouting — design wireframe and canvas structure for all 4 views
9. Charts Development and Formatting — build and style all visuals consistently
10. Dashboard / Report Development — assemble all pages with slicers, filters, and navigation
11. Insights Generation — write descriptive analysis and key takeaways for stakeholders

10. Acceptance Criteria

The dashboard will be considered complete and accepted when the following conditions are met:

- All 4 dashboard views are fully functional and render correctly in Power BI Desktop
- All 4 KPI cards display correct values with working area sparklines
- Month Name and Year slicers filter all visuals correctly on Monthly View
- Date range slicer works correctly on Consolidated View
- Patient Details grid displays all 9 required fields with accurate data
- Key Takeaways page contains complete descriptive analysis with insights
- All DAX measures return correct calculated values validated against source data
- Dashboard colour theme is consistent throughout all pages
- Data source button links correctly to the Excel file on OneDrive/Google Drive
- Document reviewed and signed off by the project stakeholder

11. Assumptions & Constraints

11.1 Assumptions

- The Excel dataset is complete, accurate, and requires no major structural changes
- A dedicated Date Table will be created in Power BI for time intelligence calculations
- Stakeholders have access to Power BI Desktop or Power BI Service to view the report
- The dataset covers only the period April 2023 – October 2024 and will not be updated retroactively
- Patient data is anonymized and compliant with applicable privacy regulations

11.2 Constraints

- No live data connection — dashboard reflects static dataset only
- Power BI Free license limitations may restrict sharing via Power BI Service
- Dataset is limited to ER records only; no outpatient or inpatient data is included
- Visuals are designed for desktop viewing; mobile layout is out of scope

12. Document Sign-Off

This BRD requires review and approval from the following stakeholders before development commences:

Name	Role	Signature	Date
	Hospital Administrator		
	Clinical Lead / ER Head		
	Operations Manager		
	Data Analyst		

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